Dhirubhai Ambani Institute of Information and Communication Technology

IT401: High Performance Computing

Lab Assignment 05 Report

Performance Analysis: Deferred vs. Immediate Replacement

Group 04
Ramani Darshit B. (202301404)
Vasu Bhesandariya (202301416)

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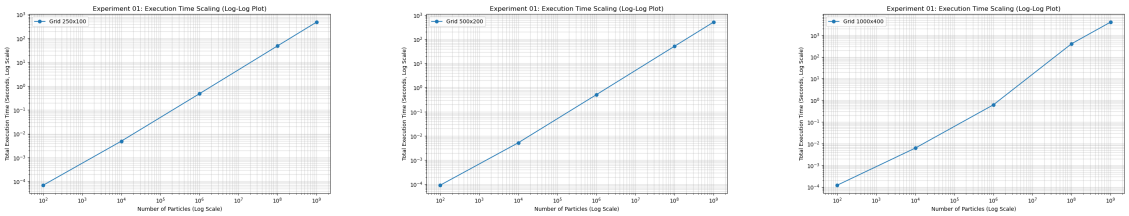
1 Approach 1: Deferred Insertion (Asgn 04)

This approach separates the movement of particles from the insertion/deletion logic. Particles exiting the boundary are handled in a sequential or separate parallel step after the main mover loop.

1.1 Experiment 01: Serial vs. Parallel (16 Threads)

Lab PC Results



HPC Cluster Results

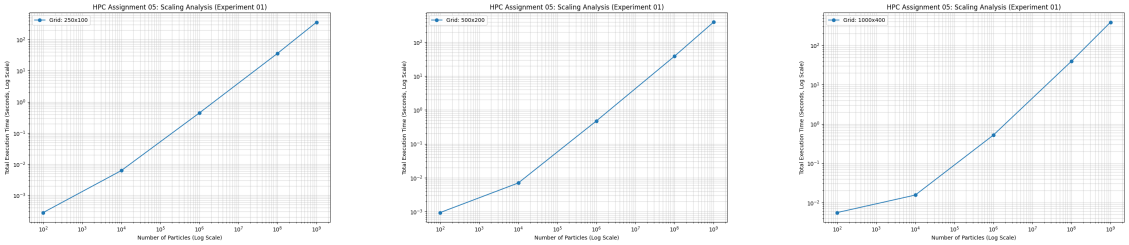


Figure 1: Approach 1 - Experiment 01: Runtime comparison (6 Plots)

1.2 Experiment 02: Scalability and Speedup (2, 4, 8, 16 Threads)

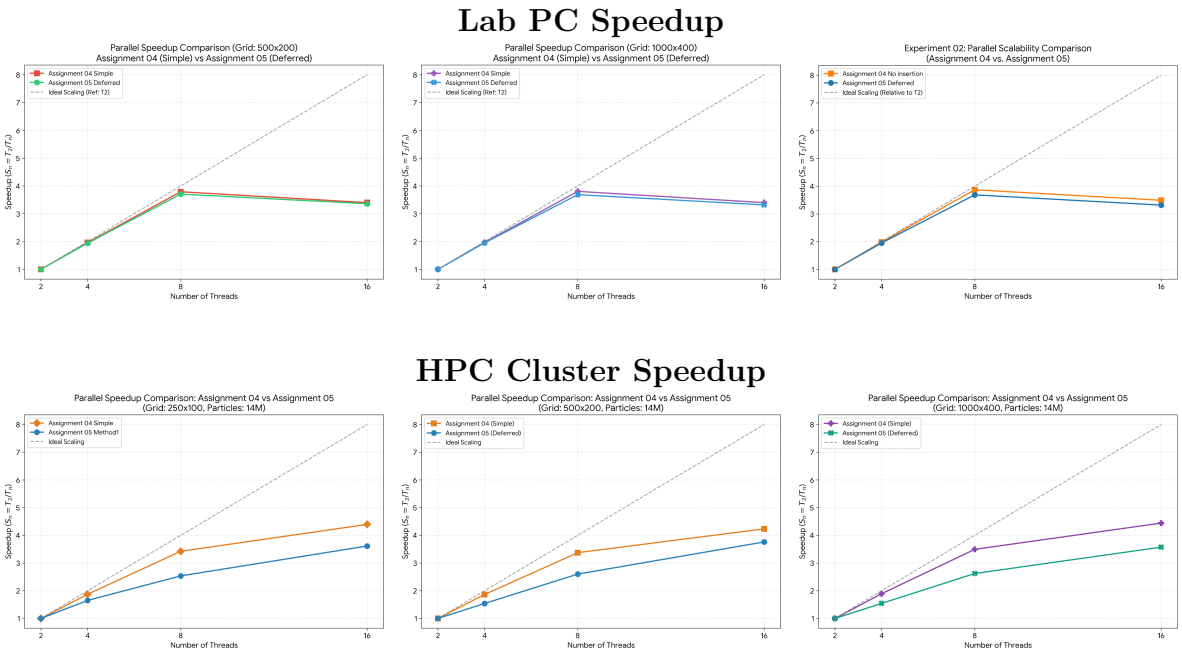


Figure 2: Approach 1 - Experiment 02: Speedup Curves (6 Plots)

2 Approach 2: Immediate Replacement (Asgn 05)

This approach resets particles to a random location immediately when a boundary violation is detected within the parallel mover loop.

2.1 Experiment 01: Serial vs. Parallel (16 Threads)

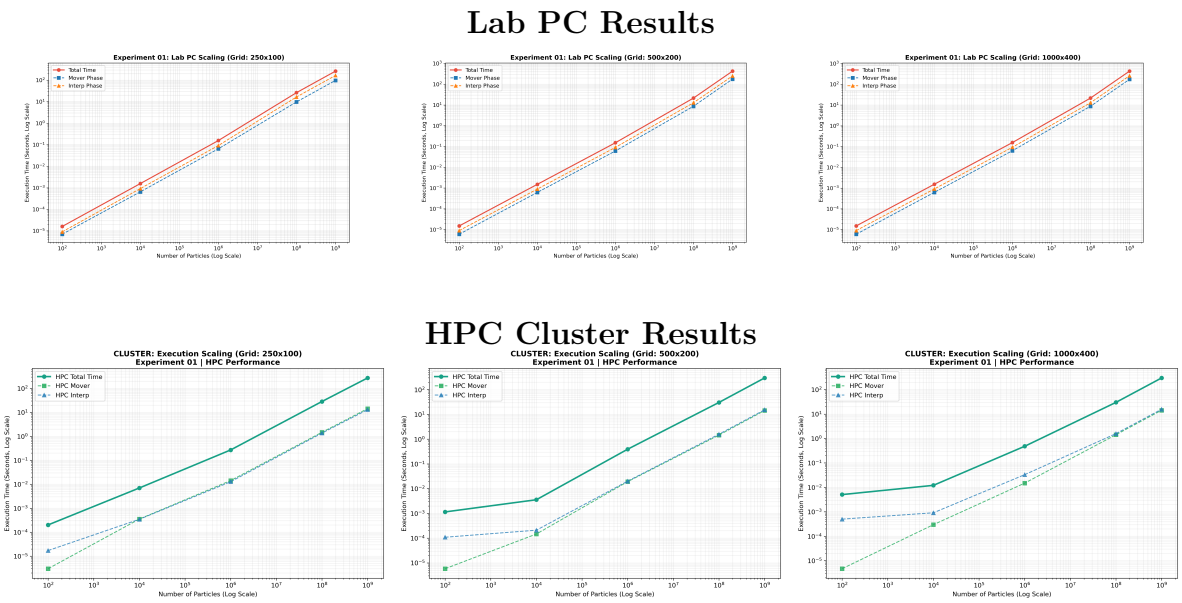
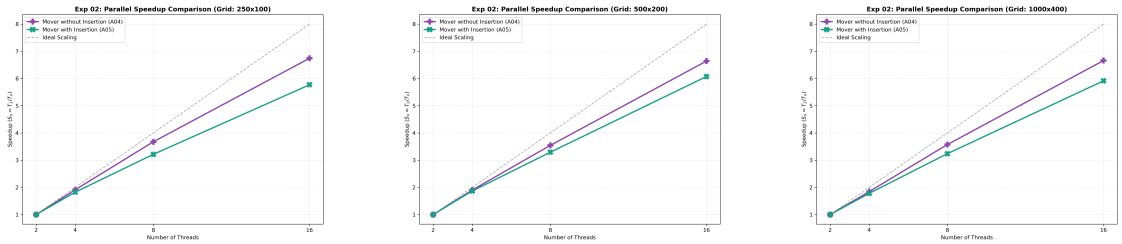


Figure 3: Approach 2 - Experiment 01: Runtime comparison (6 Plots)

2.2 Experiment 02: Scalability and Speedup (2, 4, 8, 16 Threads)

Lab PC Speedup



HPC Cluster Speedup

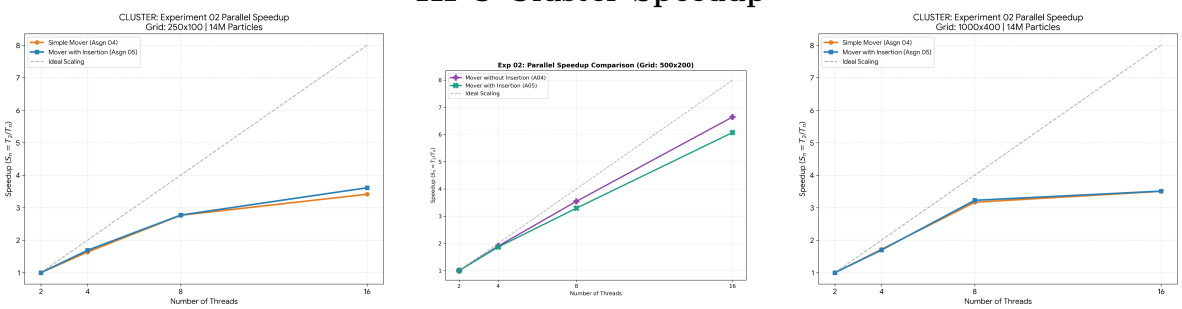


Figure 4: Approach 2 - Experiment 02: Speedup Curves (6 Plots)

3 Analysis and Conclusion

In **Approach 1**, the separation of tasks leads to cleaner parallel loops but introduces memory synchronization overhead. In **Approach 2**, the use of ‘thread$_{l}ocal$‘$randomenginesallowsforbet$ due to its advanced memory architecture.

4 External Resources

The source code, scripts, and raw data used for this assignment are available at the following link:

https://github.com/ramani3311/HPC_LABS